Market Research and Analysis

Lecture 6: AI Agents for Economic Research

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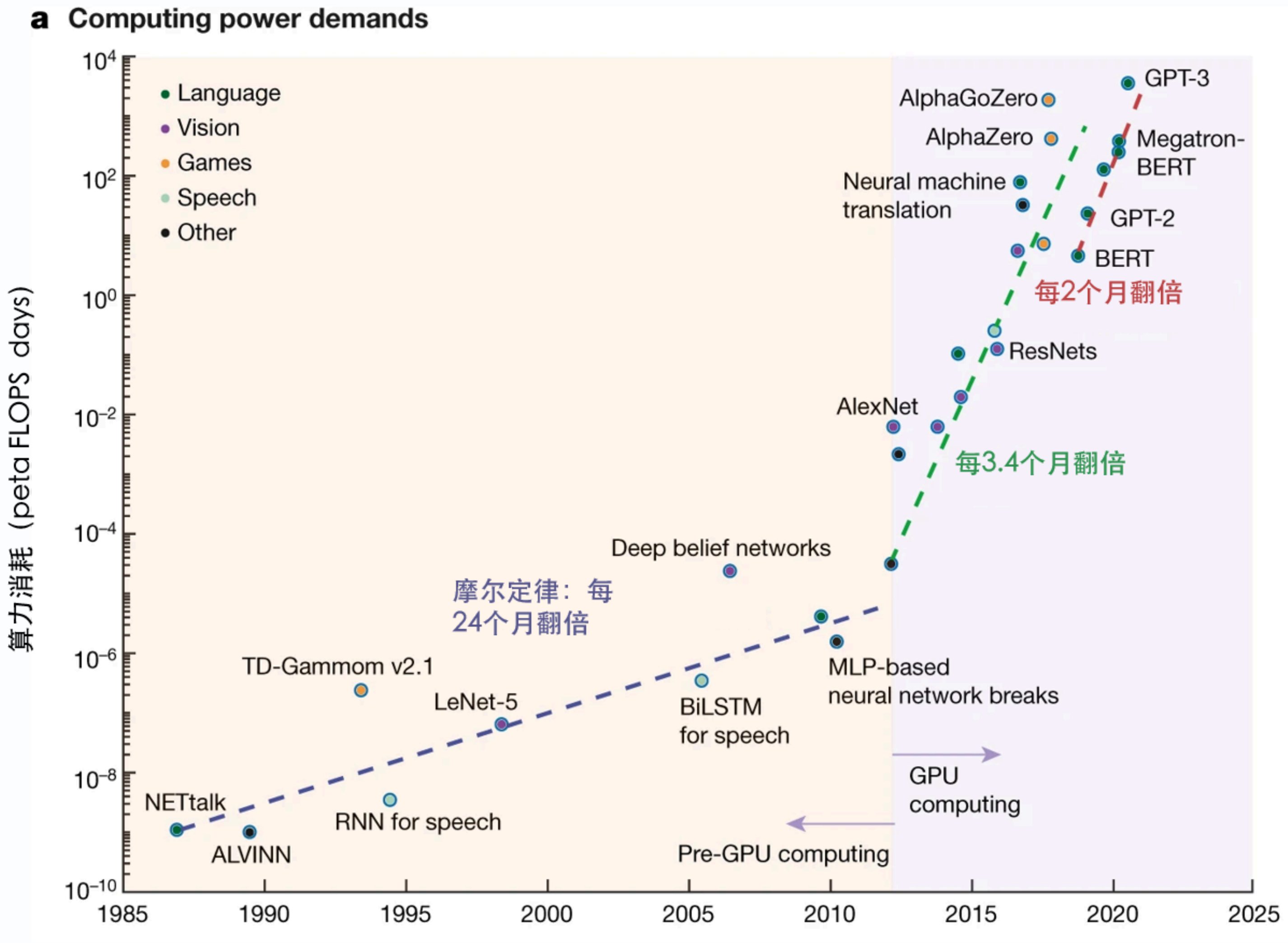
01

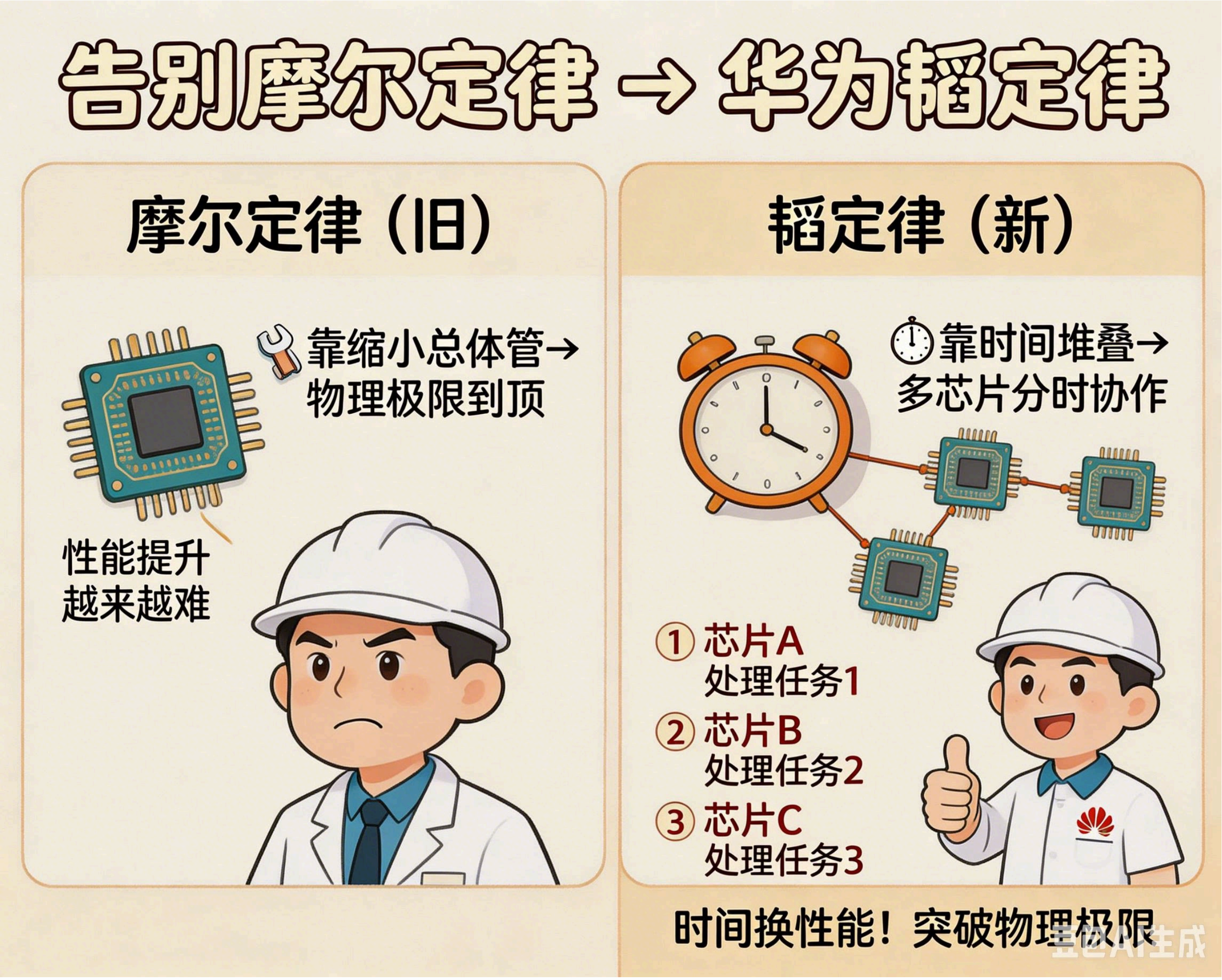
算力驱动的未来

顺应数字技术和人工智能发展大势，《中华人民共和国国民经济和社会发展第十五个五年规划纲要》将提升数智化发展水平单独成篇，聚焦算力、算法、数据高效供给和数智技术赋能经济社会发展，为未来5年数字中国建设划定清晰路线。


a Computing power demands
Language
Vision
Games
Speech
Other
算力消耗（peta FLOPS days）
GPT-3
AlphaGoZero
AlphaZero
Megatron-BERT
Neural machine translation
GPT-2
BERT
每2个月翻倍
ResNets
AlexNet
每3.4个月翻倍
Deep belief networks
摩尔定律：每24个月翻倍
TD-Gammom v2.1
LeNet-5
BiLSTM for speech
MLP-based neural network breaks
GPU computing
NETtalk
ALVINN
RNN for speech
Pre-GPU computing
1985
1990
1995
2000
2005
2010
2015
2020
2025

Moore's Law is the famous prognostication by Intel co-founder Gordon Moore that the number of transistors on a microchip would double every year or two. This prediction has mostly been met or exceeded since the 1970s — computing power doubles about every two years, while better and faster microchips become less expensive.

This rapid growth in computing power has fueled innovation for decades, yet in the early 21st century researchers began to sound alarm bells that Moore's Law was slowing down. With standard silicon technology, there are physical limits to how small transistors can get and how many can be squeezed onto an affordable microchip.


告别摩尔定律 → 华为韬定律
摩尔定律（旧）
靠缩小总体管→
物理极限到顶
性能提升
越来越难
韬定律（新）
靠时间堆叠→
多芯片分时协作
① 芯片A
处理任务1
② 芯片B
处理任务2
③ 芯片C
处理任务3
时间换性能！突破物理极限

2026年5月25日，在电气电子工程师学会（IEEE）举办的国际电路系统研讨会ISCAS 2026上，华为何庭波发表题为“半导体新路径探索与实践”的主旨演讲，发表了指导半导体产业发展的新原则——韬(τ)定律。韬(τ)定律提出以“时间(τ)缩微”替代“几何缩微”作为半导体与电子系统演进的新指导原则——通过逻辑折叠等创新技术，持续压缩信号传播时延，不断提升晶体管密度，从而实现半导体与电子系统的持续演进。

「一张图看懂摩尔定律与韬定律」

「核心思想、技术路径与产业意义对比」

「摩尔定律：几何缩微（做小）」 「两条路线，解决不同增长问题」 「韬定律：时间缩微（压快）」

「维度」	「摩尔定律」	「韬定律」
1. 「提出时间」	「1965年」	「2026年」
2. 「提出者」	「戈登·摩尔（美国）」	「何庭波（中国）」
3. 「核心思路」	「几何缩微（做小）」	「时间缩微（压快）」
4. 「优化目标」	「晶体管密度」	「信号传播时间常数 τ」
5. 「技术手段」	「光刻工艺进步」	「逻辑折叠 + 3D堆叠 + 先进封装」
6. 「迭代节奏」	「统一节奏：2年/代」	「场景驱动：1.3x~10x/年」
7. 「物理瓶颈」	「7nm以下严重收窄」	「未到理论极限」
8. 「工具链」	「EDA成熟完善」	「需全新EDA工具链」
9. 「成本趋势」	「先进制程成本飙升」	「成熟制程 + 立体设计」
10. 「产业基础」	「依赖尖端光刻设备」	「依赖先进封装能力」
11. 「中国企业影响」	「受制于设备限制」	「绕开光刻瓶颈」

「核心启示」

- 「摩尔定律偏向制程驱动」
- 「韬定律偏向架构驱动」
- 「前者依赖光刻，后者依赖封装」
- 「对中国企业具有重要战略意义」

「小白怎么理解」

「摩尔定律」

「核心是把晶体管做得更小、更密，让芯片越来越强」

VS

「韬定律」

「核心是不必一味缩小，而是通过架构、堆叠和封装把速度压快」

「一个主要追求做小，一个更强调跑快」

一张图看懂普及摩尔定律和韬定律两条定律的全面对比和小白解释

关键区别

- 摩尔定律关注硬件性能增长
- 韬定律关注系统智能增长
- 一个偏底层算力，一个偏上层能力

核心思想与技术路径

摩尔定律：靠芯片变强　　**韬定律：靠系统变聪明**

摩尔定律

小白理解：让芯片上的晶体管越来越多，算力越来越强

1 核心思想

通过提升单位芯片上的晶体管数量，持续推动计算性能提升与成本下降

2 技术路径

制程缩小 → 晶体管密度提升 → 芯片性能增强 → 单位算力成本下降

3 典型特征

- 摩件驱动
- 依赖半导体工艺进步
- 增长来自底层器件迭代

摩尔定律	比较维度	韬定律
芯片算力	提升对象	任务智能
半导体工艺	主要抓手	模型+数据+工具+系统
器件密度提升	增长来源	协同优化与流程放大
更快、更便宜的计算	结果表现	更强、更完整的任务完成能力

韬定律

小白理解：不只是换更强芯片，而是让模型、数据、系统和工具一起变强

1 核心思想

通过模型、数据、工具、推理、工作流与系统协同优化，
持续放大智能产出与任务完成能力

2 技术路径

更强模型 — 更多高质量数据 — 更长上下文 — 工具调用 — 工作流编排 — 人机协同优化

- 系统编排与协同层
- 推理引擎与执行层
- 算力与基础设施层

3 典型特征

- 系统驱动
- 依赖软件与协同能力提升
- 增长来自整体智能系统迭代

小白版一句话

摩尔定律
核心是把机器本身做得更强

韬定律
核心是让整套智能系统更会做事

★ 前者更像升级发动机，后者更像升级整辆车和驾驶系统 ★

02

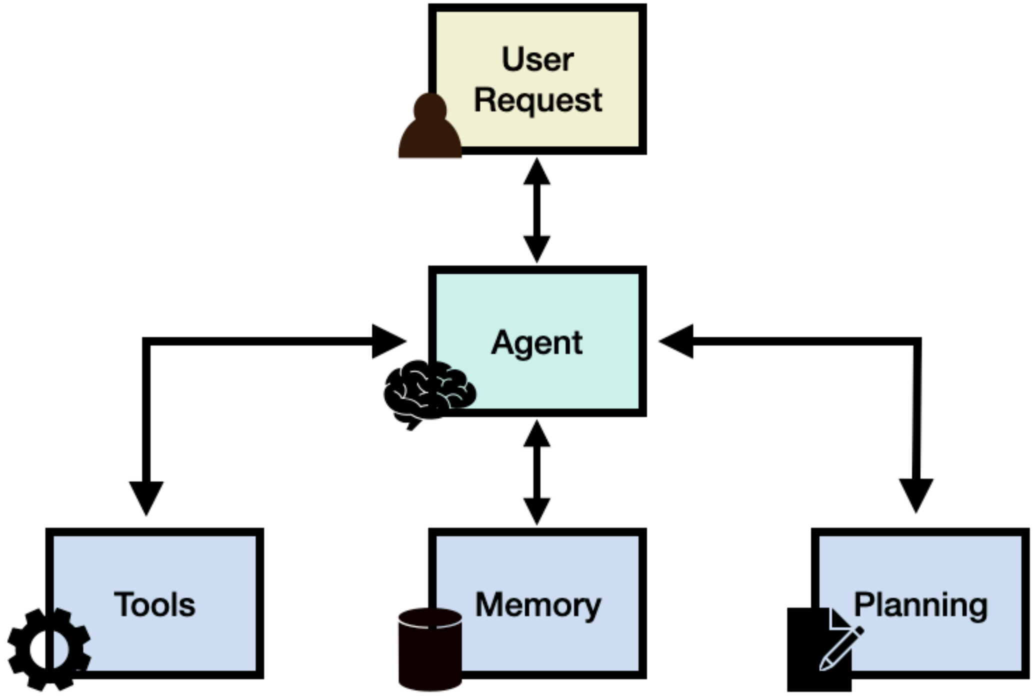
AI Agent

> AI is no longer only a writing assistant. It is becoming a research partner that extends your second brain.

Agent 就是一个能干活的智能助手。

Agent = LLM (大脑) + Planning (规划) + Tool use (执行) + Memory (记忆)

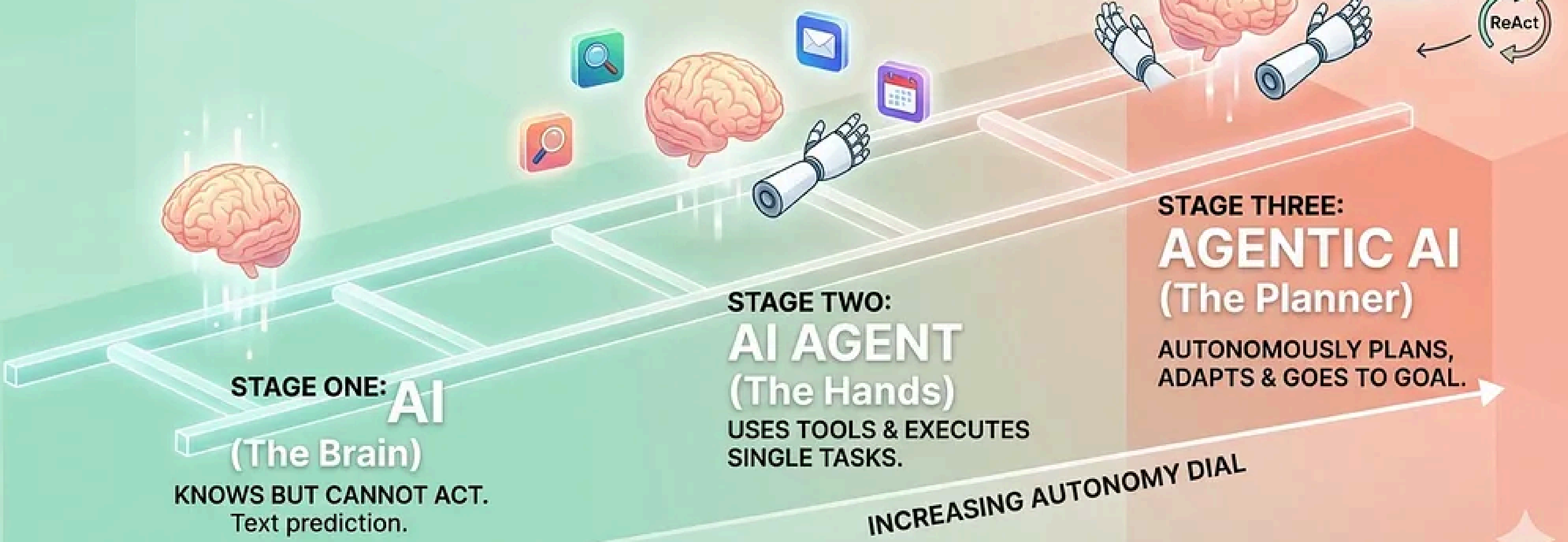
学习 Agent 需要思维转变： 从 `对话框问答` 进化为 `目标驱动的任务执行` 。



它就像是一位**拥有“数字大脑”和“手脚”的私人助理**，不仅能回答问题，还能替你接管复杂的任务并执行操作：

AI, AI Agents, and Agentic AI, Explained All the Way Down

The Brain, the Hands, and the Party Planner

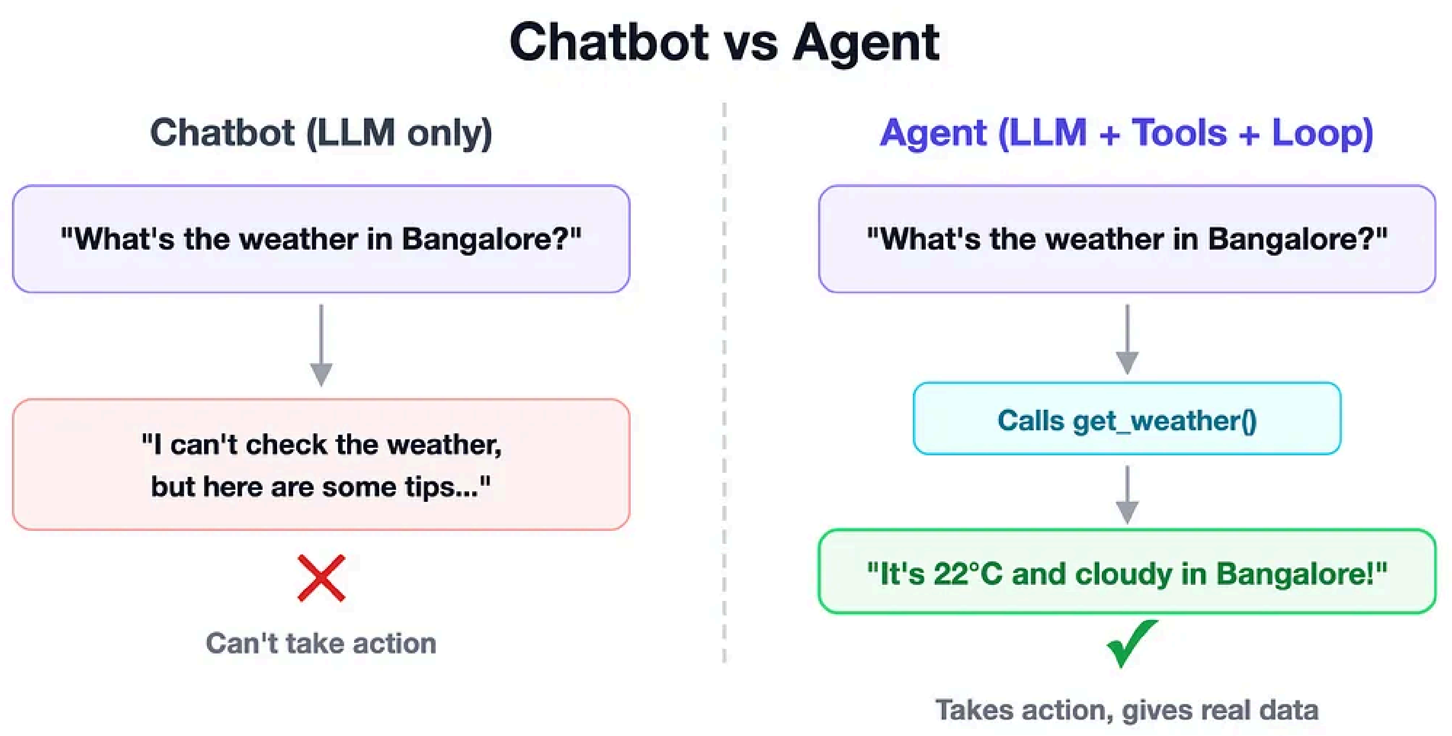


INCREASING AUTONOMY DIAL

AI 智能体 vs. 传统 AI（聊天机器人）

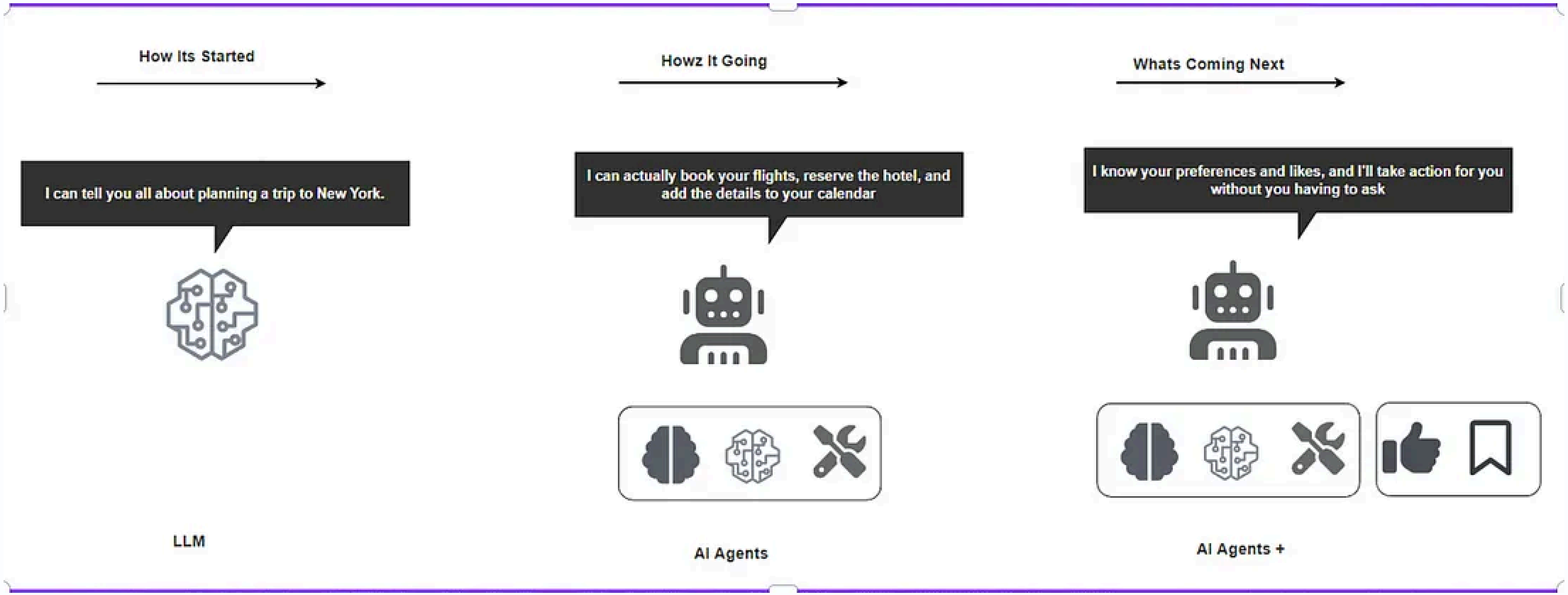
传统的聊天机器人如 ChatGPT 就像简单的客服只能**被动响应**你的输入，需要人类给出详细的每步指示；而 AI Agent 是**目标驱动**的，一旦你设定了目标，它能自主出击，独立完成多步骤操作。例如：

- 普通聊天机器人：你问“北京天气如何？”，它会回答“今天北京下雨”。
- AI Agent：你下达指令“帮我查一下下周去北京的机票和酒店，并在预算内完成预订”。Agent 会自主查询多个网站（工具）、比价（规划）、最终生成订单（行动），全程无需人类干预。



AI Agent 更像一个有自主性的员工，它能够：

- **理解任务目标**：明白你想要什么结果
- **制定计划**：思考如何达成目标
- **使用工具**：调用各种资源和API
- **自我调整**：根据反馈优化策略
- **持续执行**：直到完成任务或遇到无法解决的问题



一个完整的 AI Agent 通常包含以下四个核心模块：

1. **大脑（大语言模型LLM）**：负责理解意图、逻辑推理和生成决策。
2. **规划（Planning）**：能将复杂的大目标拆解成一步步可执行的小任务，并在遇到阻碍时自我反思与调整。
3. **记忆（Memory）**：记录过去的交互历史（短期记忆）和存储专业知识库（长期记忆）。
4. **工具使用（Tool Use）**：调用各种外部工具（如搜索引擎、数据库、计算器、API 等）来完成具体动作。

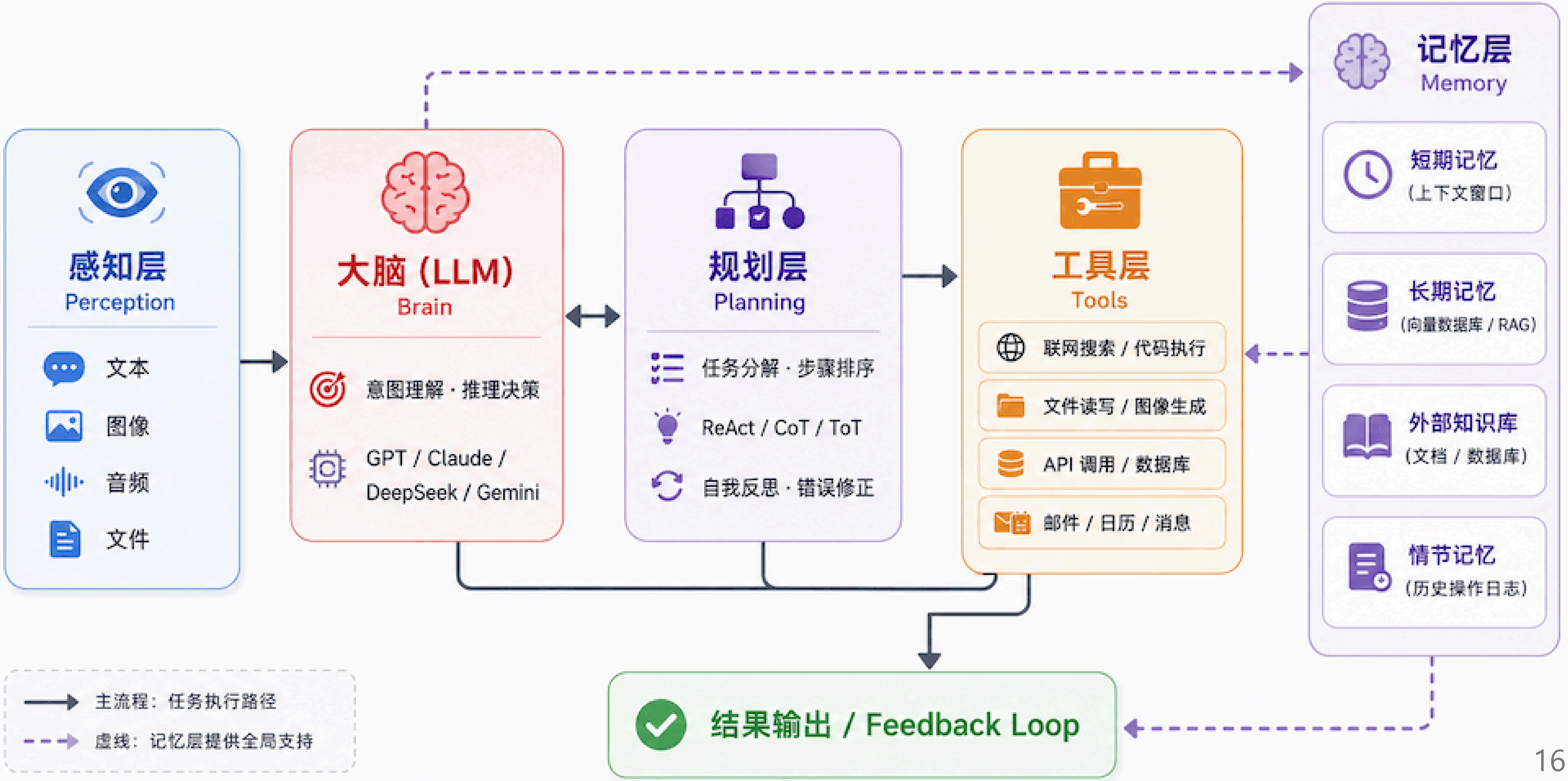
与普通大模型的差异点:

- 普通大模型：生成文本、图片等
- Agent：生成行动并执行行动，能完成实际工作

Agent 与传统 AI 模型的区别

维度	传统 AI 模型	AI Agent
交互方式	单次输入输出	多轮对话、持续交互
决策能力	基于输入直接推理	规划、反思、迭代优化
工具使用	无法主动调用外部工具	可调用搜索、计算器、API 等
记忆机制	仅限当前上下文	短期+长期记忆
目标导向	完成单一预测任务	完成复杂目标
错误处理	输出即结束	可自我纠错、重试

AI Agent 五大核心组件



0、感知层 (Perception) —— 餐厅的前台

角色：负责接待顾客，理解来自外部世界的所有输入。

Agent 在行动之前，必须先"看到"和"听到"外部信息。现代 Agent 已经不限于纯文本输入，而是具备**多模态感知**能力：

- **文本输入**：用户的自然语言指令、文档内容、代码。
- **图像 / 视频**：截图、图表，Agent 可以直接"看图"理解。
- **结构化数据**：表格、JSON、数据库查询结果。
- **环境状态**：在计算机操作类 Agent 中，当前屏幕状态、网页结构等。
- **工具返回结果**：上一步工具调用的输出，会作为新的感知输入进入下一轮循环。

感知层的输入经过整合，形成 Agent 的"当前上下文"，送入大脑进行理解和决策。

1、大脑 (Brain) —— 也就是大模型

角色：餐厅的**主厨兼经理**。

这是 Agent 最核心的部分（比如 GPT-4、Claude、DeepSeek、通义千问）。

- 它负责**听懂**你想吃什么（理解意图）。
- 它负责**指挥**其他人干活（决策）。
- 如果没有它，整个餐厅就瘫痪了。

大脑做的三件核心事

意图理解	解析用户输入，明确目标是什么	听懂顾客点了什么
推理决策	综合上下文和记忆，判断下一步该做什么	主厨决定先处理哪道菜
工具调用判断	判断是否需要调用外部工具，选择哪个工具、传入什么参数	决定用哪口锅、让谁去买食材

大脑的"智力天花板"决定了整个 Agent 的上限。同一套工具和规划框架，接入能力更强的基础模型，任务完成质量往往有质的飞跃。

2、工具 (Tools) —— 厨房里的设备

角色：**厨具和帮手**。

光有主厨（大脑）是不够的，还得有锅碗瓢盆才能做菜。对于 AI Agent 来说，工具就是能把决策转化为真实动作的执行单元。

工具可以按照用途分为四大类：

类别	常见工具	作用
信息获取	联网搜索、网页抓取、文档读取、数据库查询	获取 Agent 自身知识之外的实时或专业信息
计算执行	代码解释器、数学计算引擎、沙箱环境	处理需要精确计算或程序逻辑的任务
内容生成	图像生成、语音合成、文档导出	产出非文本形式的内容
系统交互	API 接口、邮件、日历、文件操作、消息发送	与外部系统、服务和真实世界进行交互

常见工具举例：**联网搜索** 信息获取像去菜市场买新鲜食材，**代码解释器** 计算执行像精密的烤箱，处理复杂计算，**画图工具** 内容生成像摆盘师，负责美观，**API 接口** 系统交互像外卖小哥，连接外部世界。

3、记忆 (Memory) —— 顾客记录本

角色：服务员的记性。

你肯定不喜欢每次去餐厅都要重新报一遍：我不吃香菜！

Agent 的记忆分为以下几种类型：

- **短期记忆（In-Context Memory）**：即当前对话的上下文窗口。记住刚才你说了啥（比如你刚点了鱼，下一句说"要微辣"，它知道是指鱼）。受限于模型的上下文长度，通常在 8K 到 200K token 之间。
- **长期记忆（External Memory）**：记住你的长期偏好（比如你是素食主义者，或者你的家庭住址）。通常通过向量数据库实现持久化存储。
- **情节记忆（Episodic Memory）**：对历史任务执行过程的记录，包括"上次遇到这种情况我是怎么处理的"，帮助 Agent 从过去的经验中学习。
- **语义记忆（Semantic Memory）**：抽象的知识和事实，通常来自预训练阶段已经内化的内容，也可通过 RAG（检索增强生成）动态补充。

4、规划 (Planning) —— 烹饪流程单

角色：**后厨的出餐 SOP**（Standard Operating Procedure，标准作业程序）。

当你点了一份佛跳墙，主厨不会乱做，而是会在脑子里生成一个清单：

1. 先备料（鲍鱼、海参...）
2. 再熬汤
3. 最后慢炖

Agent 也是一样。当你给它一个复杂任务（比如"写一份竞品分析报告"），它会自己拆解：

- 第一步：去搜集竞品 A、B、C 的资料。
- 第二步：对比它们的价格和功能。
- 第三步：把对比结果写成文章。
- 第四步：检查一遍有没有错别字。

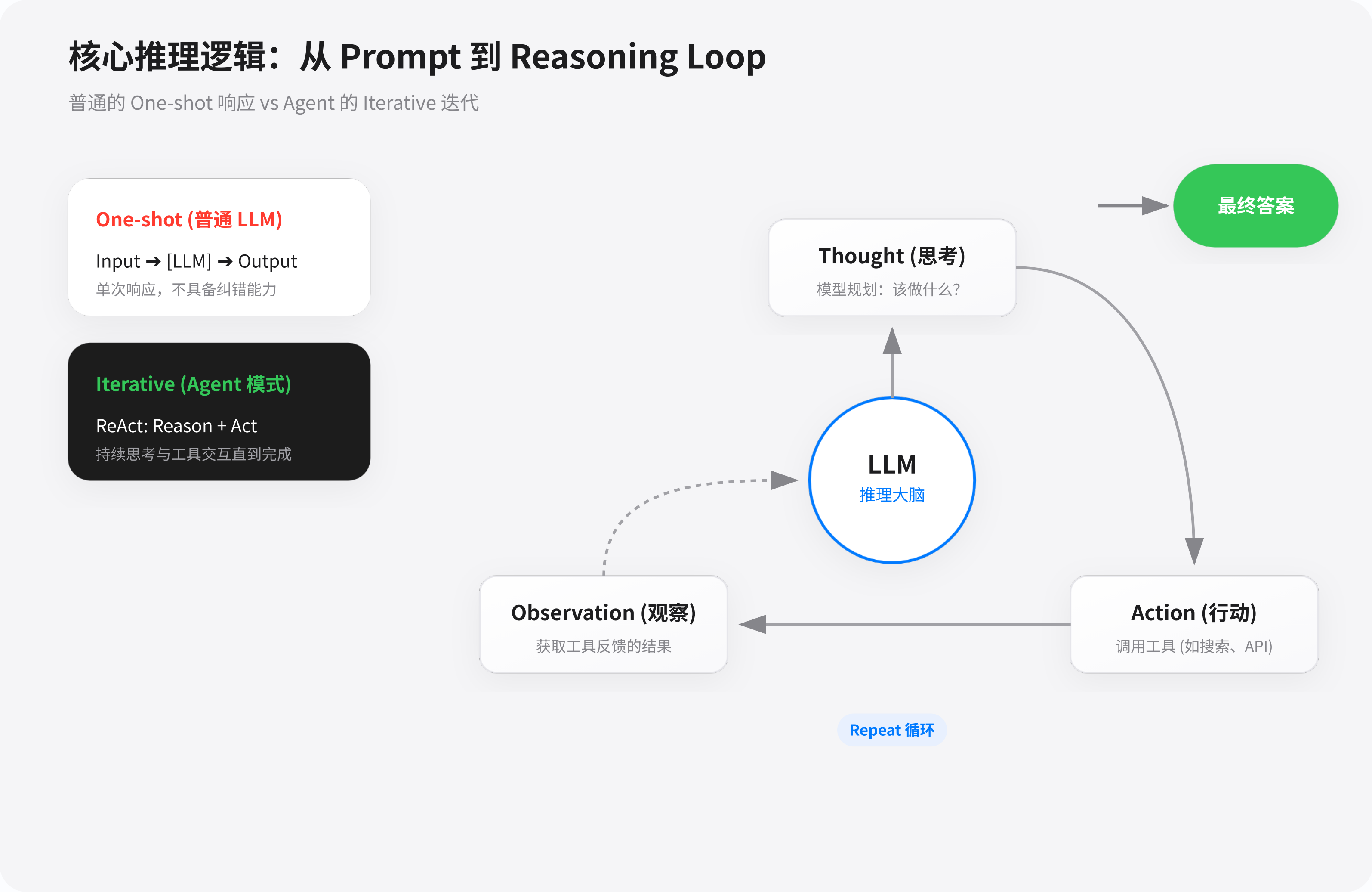
5、Agent 运行循环 (Agent Loop)

以上各组件并非孤立存在，它们组成一个持续迭代的**感知—思考—行动—观察**闭环，这就是"Agent Loop"。Agent 不断重复这个循环，直到任务完成或达到终止条件。

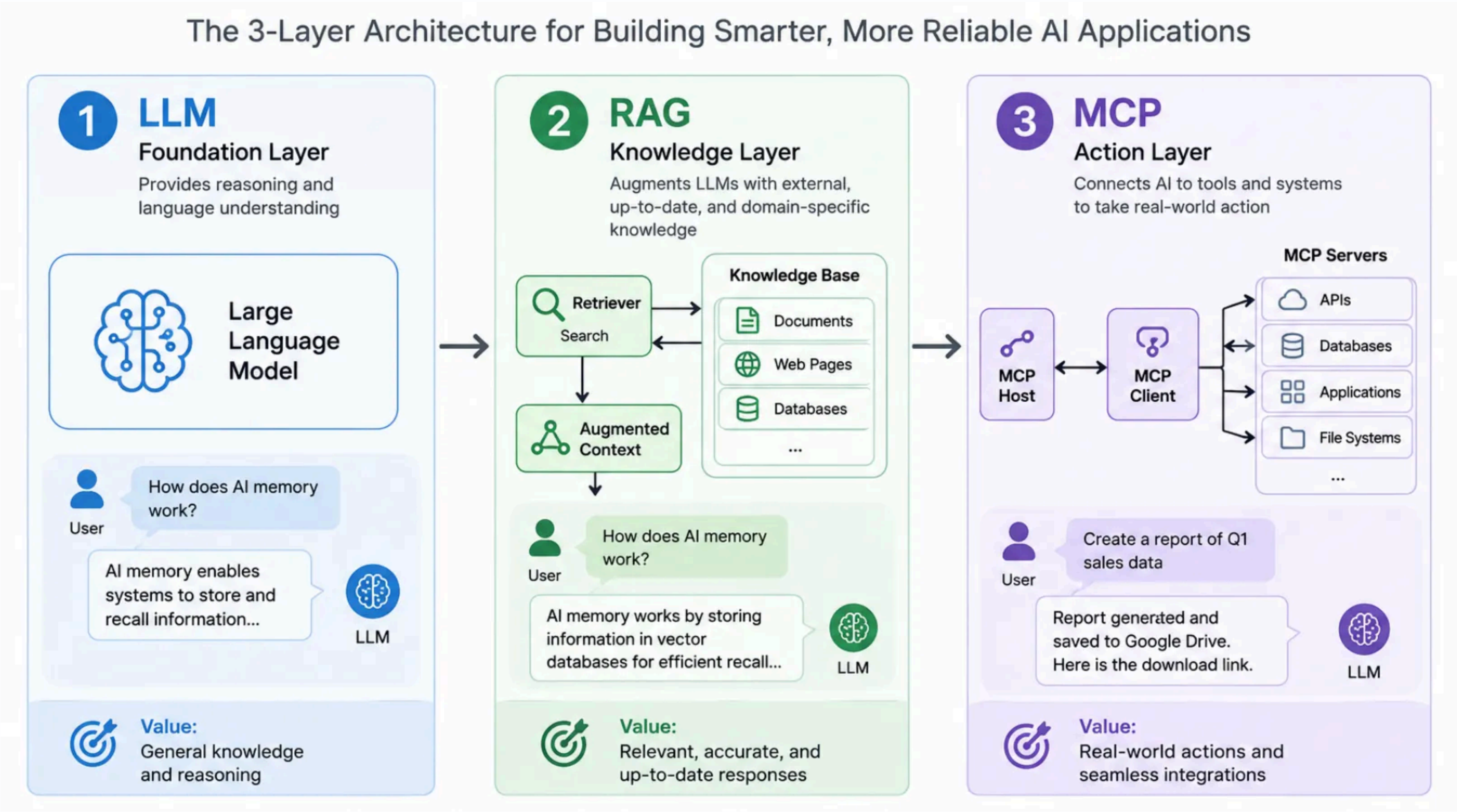
感知（接收输入/环境状态）→ 思考（LLM 推理/规划分解）→ 行动（调用工具/执行操作）→ 观察（获取结果/更新记忆）→ 达到终止条件（任务完成）/继续循环（任务未完成）

这个循环让 Agent 具备了**在失败时自我纠错**的能力：如果某一步工具调用返回了错误或意外结果，"观察"阶段会将这个信息反馈给大脑，大脑在下一轮"思考"时就会调整策略。

普通的 LLM 只是 One-shot（一次性） 的响应，而 Agent 的核心在于 Iterative（迭代）。
ReAct 模式 (Reason + Act) 是目前最主流的 Agent 推理逻辑：



从基础模型到智能体的完整AI系统架构

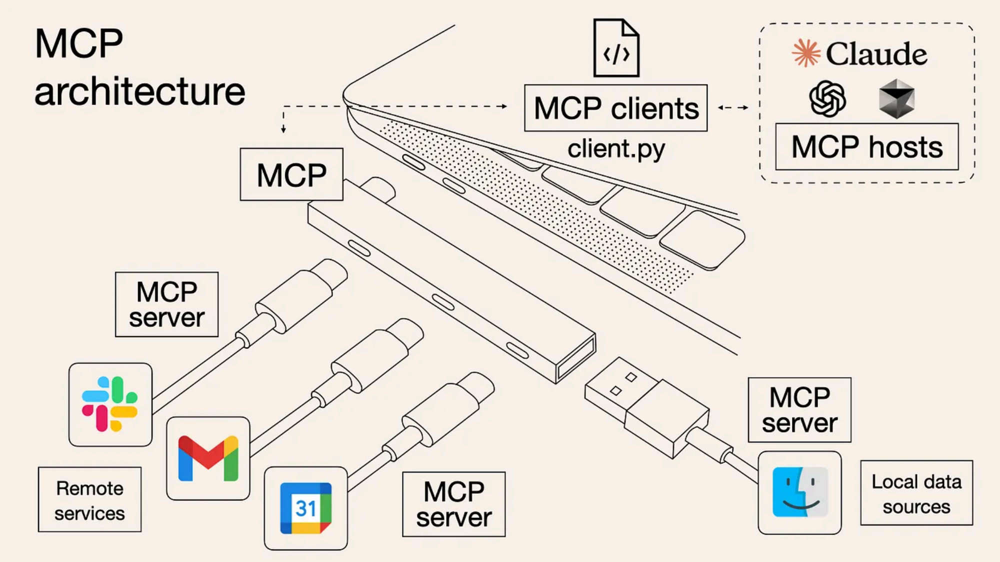

The 3-Layer Architecture for Building Smarter, More Reliable AI Applications
1 LLM
Foundation Layer
Provides reasoning and language understanding
Large Language Model
User
How does AI memory work?
AI memory enables systems to store and recall information...
LLM
Value:
General knowledge and reasoning
2 RAG
Knowledge Layer
Augments LLMs with external, up-to-date, and domain-specific knowledge
Retriever
Search
Knowledge Base
Documents
Web Pages
Databases
...
Augmented Context
User
How does AI memory work?
AI memory works by storing information in vector databases for efficient recall...
LLM
Value:
Relevant, accurate, and up-to-date responses
3 MCP
Action Layer
Connects AI to tools and systems to take real-world action
MCP Servers
MCP Host
MCP Client
APIs
Databases
Applications
File Systems
...
User
Create a report of Q1 sales data
Report generated and saved to Google Drive. Here is the download link.
LLM
Value:
Real-world actions and seamless integrations

What is the Model Context Protocol (MCP)?

MCP (Model Context Protocol) is an open-source standard for connecting AI applications to external systems.

Using MCP, AI applications like Claude or ChatGPT can connect to data sources (e.g. local files, databases), tools (e.g. search engines, calculators) and workflows (e.g. specialized prompts)—enabling them to access key information and perform tasks. `Zotero MCP`, `Stata MCP`

Think of MCP like a USB-C port for AI applications. Just as USB-C provides a standardized way to connect electronic devices, MCP provides a standardized way to connect AI applications to external systems.



What are Agent Skills?

Agent Skills are a lightweight, open format for extending AI agent capabilities with specialized knowledge and workflows.

At its core, a skill is a folder containing a `SKILL.md` file. This file includes metadata (`name` and `description`, at minimum) and instructions that tell an agent how to perform a specific task. Skills can also bundle scripts, reference materials, templates, and other resources.

```
my-skill/
├── SKILL.md          # Required: metadata + instructions
├── scripts/          # Optional: executable code
├── references/       # Optional: documentation
├── assets/           # Optional: templates, resources
└── ...               # Any additional files or directories
```

GitHub - meleantonio/awesome-econ-ai-stuff: AI Skills for Economists · GitHub

Academic Research Skills

The Claude Code Suite for Researchers

Deep Research
13-agent literature team
Semantic Scholar · PRISMA

Academic Paper
12-agent writing system
Style calibration · LaTeX

Claude Code

Reviewer
Read-only peer review
R&R traceability matrix

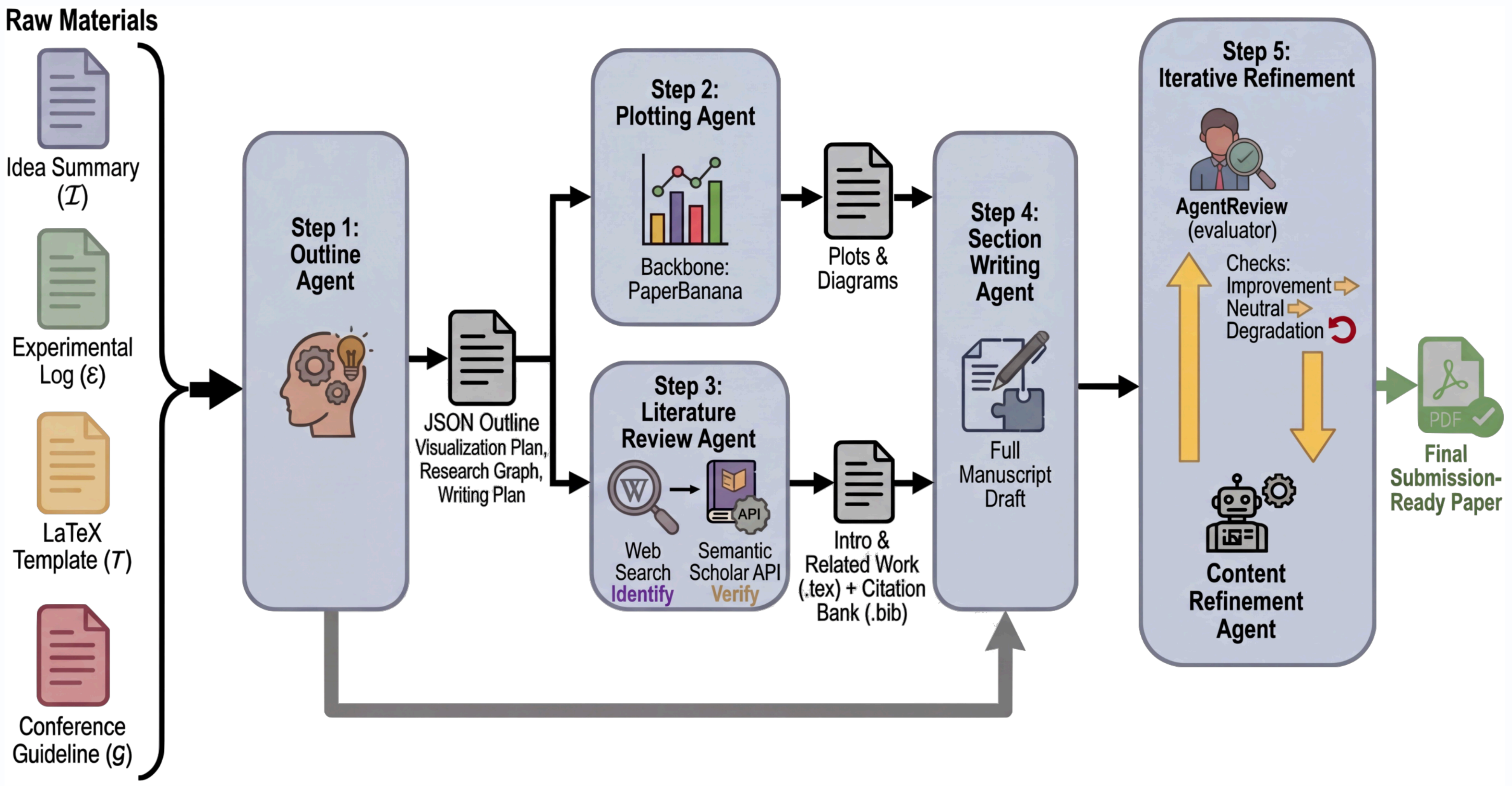
Pipeline
10-stage automation
Research → review → export

COPILOT, NOT THE PILOT

Subagent and Multi-Agents

- **agent** – one autonomous actor that takes your goal and runs with it end to end
- **subagents** - a setup where a main agent acts as orchestrator, delegating work to other agents it controls. The main agent owns the flow, order, and coordination.
- **multi agents** – two or more main agents, each acting independently but able to collaborate, negotiate, or exchange results. No single agent is "the boss".

These are just different ways to get stuff done with AI. Kind of like deciding if you want to work solo, pair program, or lead a squad.



Raw Materials
Idea Summary ($\mathcal{I}$)
Experimental Log ($\mathcal{E}$)
LaTeX Template ($\mathcal{T}$)
Conference Guideline ($\mathcal{G}$)
Step 1: Outline Agent
JSON Outline
Visualization Plan, Research Graph, Writing Plan
Step 2: Plotting Agent
Backbone: PaperBanana
Plots & Diagrams
Step 3: Literature Review Agent
Web Search
Identify
API
Semantic Scholar API
Verify
Intro & Related Work (.tex) + Citation Bank (.bib)
Step 4: Section Writing Agent
Full Manuscript Draft
Step 5: Iterative Refinement
AgentReview (evaluator)
Checks:
Improvement
Neutral
Degradation
Content Refinement Agent
PDF
Final Submission-Ready Paper

What is Git?

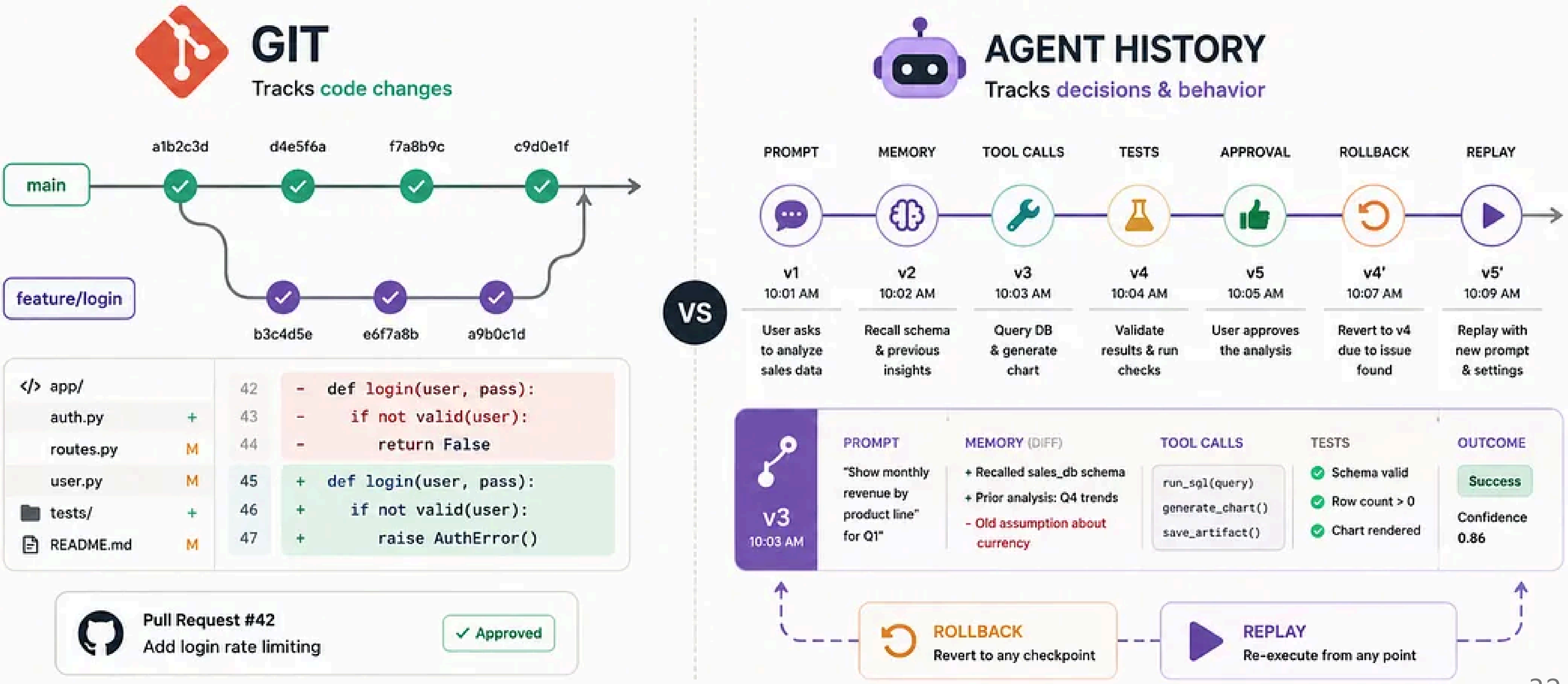
Git is a free, open-source distributed **version control system** designed to track changes in source code during software development. It allows multiple developers to work on the same project simultaneously without overwriting each other's work. Rather than storing full duplicates of your project folders, Git takes "snapshots" of your files over time, building a timeline you can easily navigate or roll back if something breaks.

Git vs. GitHub (They Are Different)

- **Git** is the local command-line software tool that manages your file history.
- **GitHub** is a cloud-based hosting platform where developers store their Git repositories to collaborate with others online. Alternative platforms include GitLab and

Git vs Agent History

CODE CHANGES vs DECISIONS & BEHAVIOR

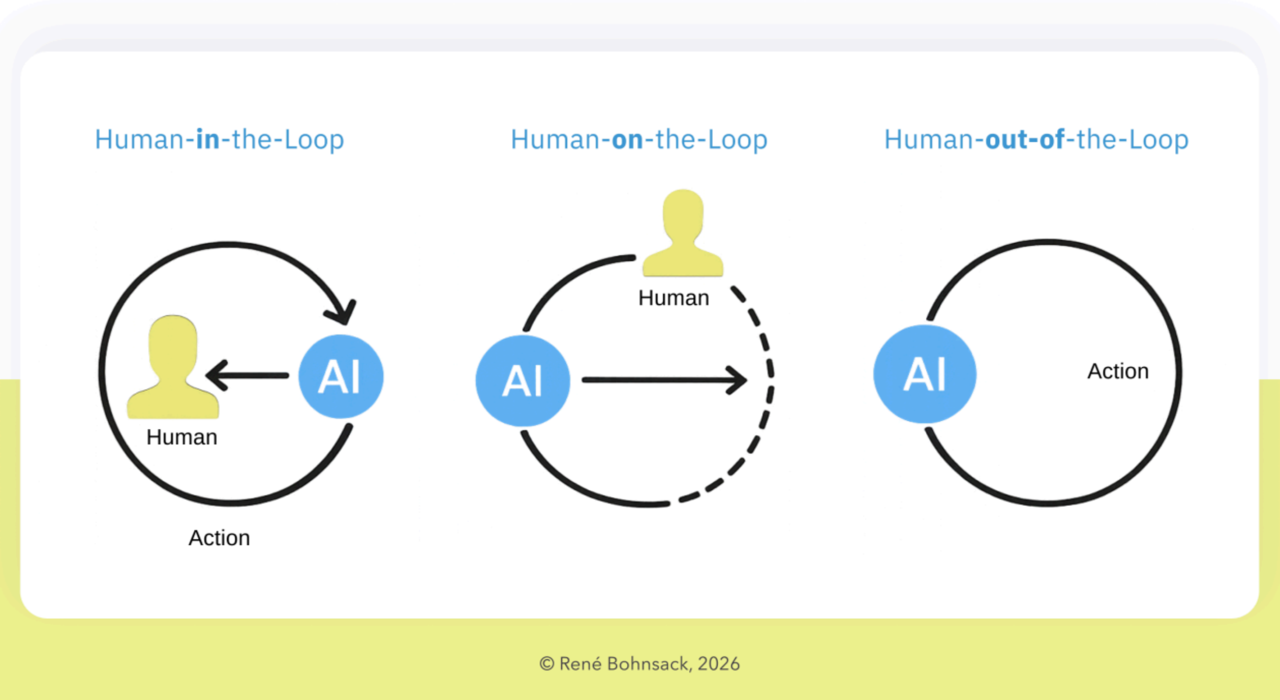


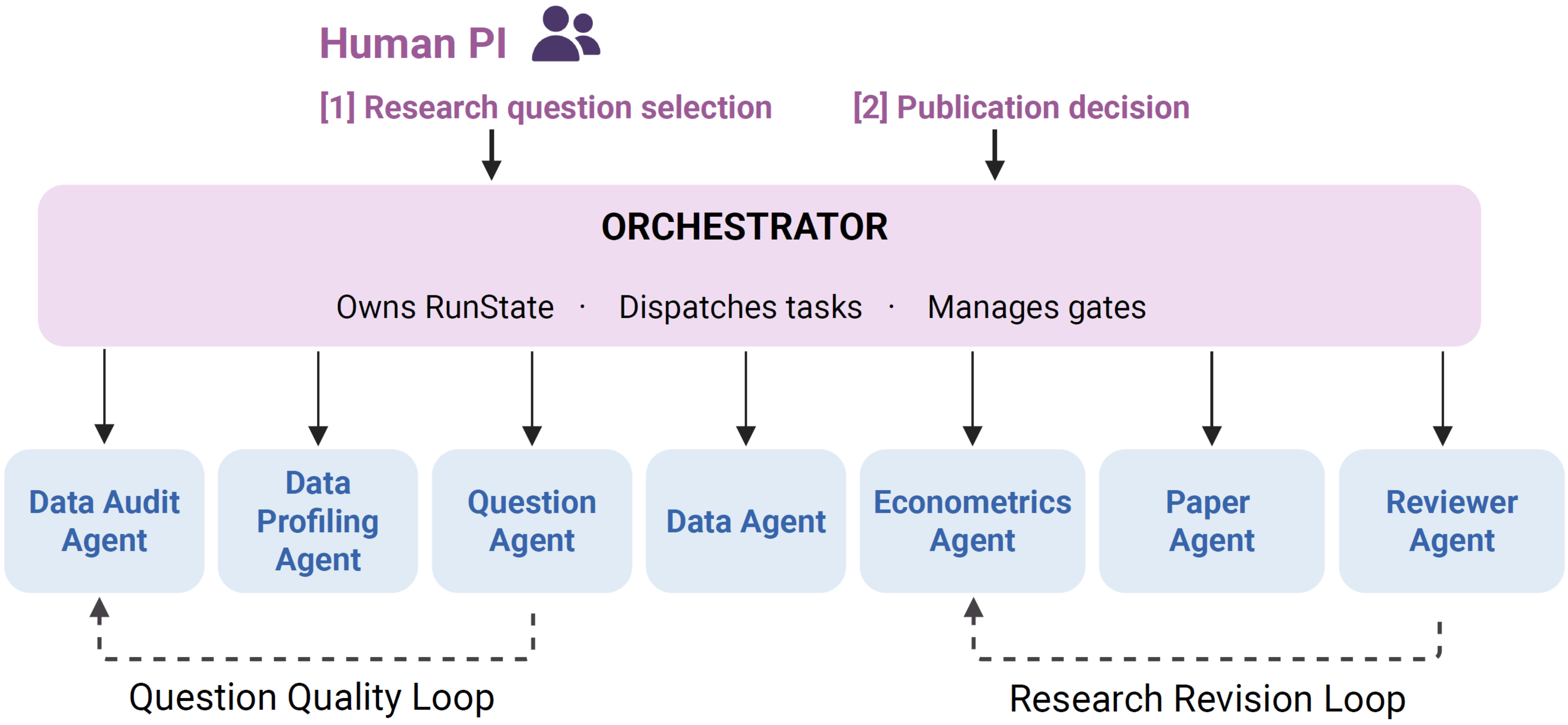
Human In the Loop

Artificial intelligence (AI) and machine learning (ML) models rely on high-quality labeled data to function effectively. Annotation software plays a key role in labeling images, text, video, and audio, making AI training possible.

While automated softwares have significantly improved efficiency, human involvement remains essential in maintaining accuracy, handling complex cases, and reducing bias in AI models.

The approach that combines automation with human expertise is called human-in-the-loop (HITL) annotation. It ensures that AI models receive precisely labeled data, leading to better decision-making and higher accuracy.





HLER: Human-in-the-Loop Economic Research via Multi-Agent Pipelines for Empirical Discovery

Hooks vs. Prompts: Why Determinism Beats Probability in AI Coding

Prompts are Probabilistic Suggestions

Instructions lose weight as context windows fill, leading to inconsistent compliance.

Probability

leaky funnel

Feature	Prompts (Suggestions)
Logic	Probabilistic (May forget)
Execution	Inside Context Window
Success Rate	~85% Compliance

85% **Compliance Problem**

Prompt-based instructions allow roughly 18 unchecked errors to slip through every two weeks.

Hooks are Deterministic Infrastructure

0101001010101010
101001
010110
0110101010000001011

Determinism

Hooks execute outside the reasoning loop, making it impossible for agents to forget.

Feature	Hooks (Enforcement)
Logic	Deterministic (Always runs)
Execution	Outside Reasoning Loop
Success Rate	100% Compliance

Compliance

Deterministic execution ensures all rules are followed without failure.

The Enforcement Mechanism

Binary Exit Code Protocol

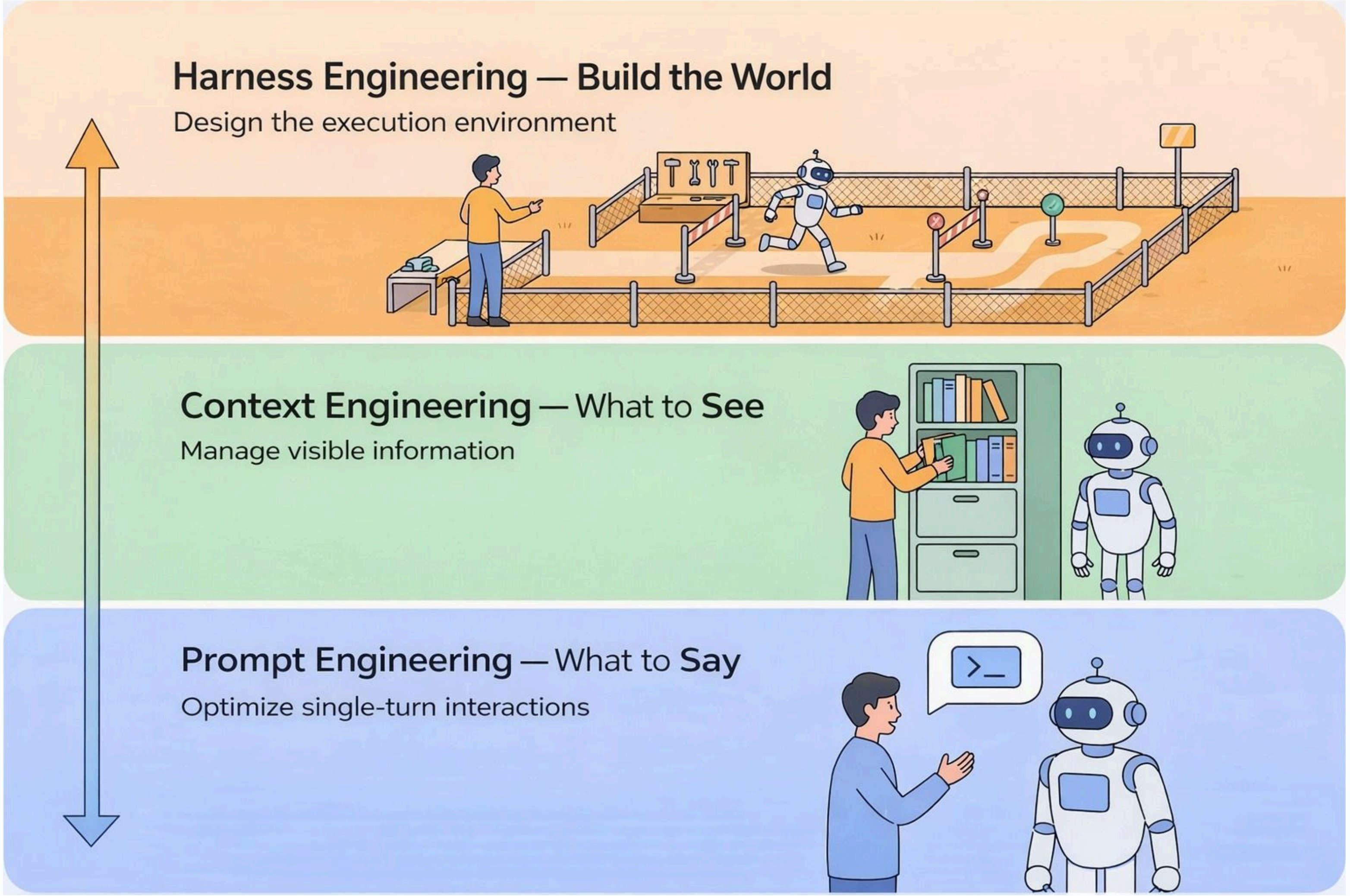
Exit Code 0 → OPERATIONS

Exit Code 2 → BLOCKS & FORCES FIXES

- **Deploy Command Hooks** (Speed)
- **Prompt Hooks** (Judgment)
- **Agent Hooks** (Audits)

"Don't Stop" Quality Gate

Use Stop hooks to prevent completion until all automated test suites pass.


Harness Engineering — Build the World
Design the execution environment
Context Engineering — What to See
Manage visible information
Prompt Engineering — What to Say
Optimize single-turn interactions
>_

AI Agents for Economic Research

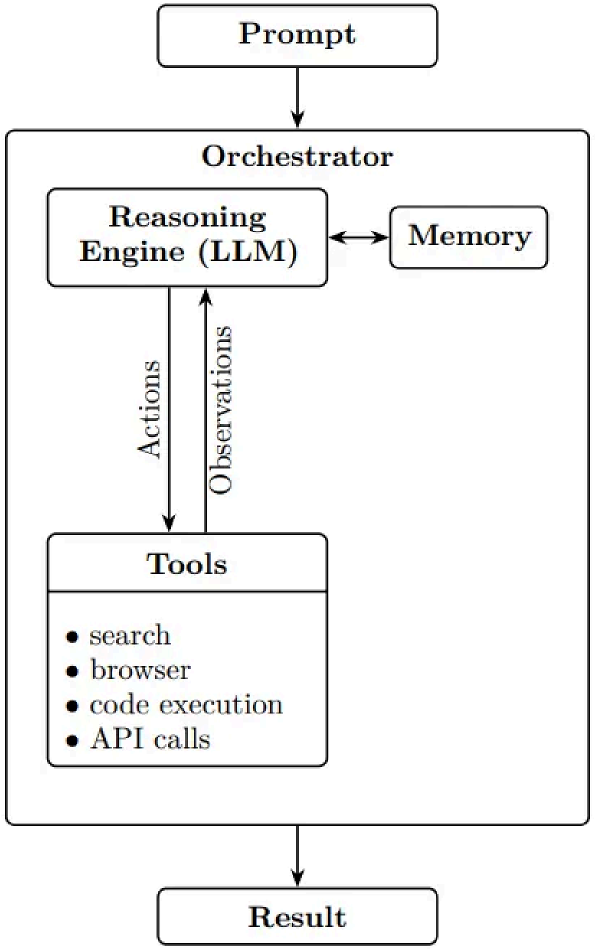


Figure 1. AI Agent Architecture

Deep Research Systems: Literature Reviews in Minutes

Deep Research agents demonstrate this capability in a powerful way. These multi-agent systems can parallel-process hundreds of sources in minutes, producing comprehensive research reports with accurate citations. When given a research question, an orchestrator agent:

1. Decomposes the question into focused subtasks
2. Spawns specialized agents to investigate different aspects in parallel
3. Synthesizes findings into coherent narratives

Deep Research systems are available in all the leading chatbots by click on "Deep Research" (see, e.g., in ChatGPT, Gemini, or Claude). While these systems compile existing knowledge rather than generating truly novel insights, they dramatically accelerate the time-intensive work of information gathering and initial synthesis. Literature reviews that once took weeks can now be completed in under an hour.

The quality of the output varies: these systems still struggle at prioritizing research at the frontier, but they are already quite capable at synthesizing well-established bodies of literature.

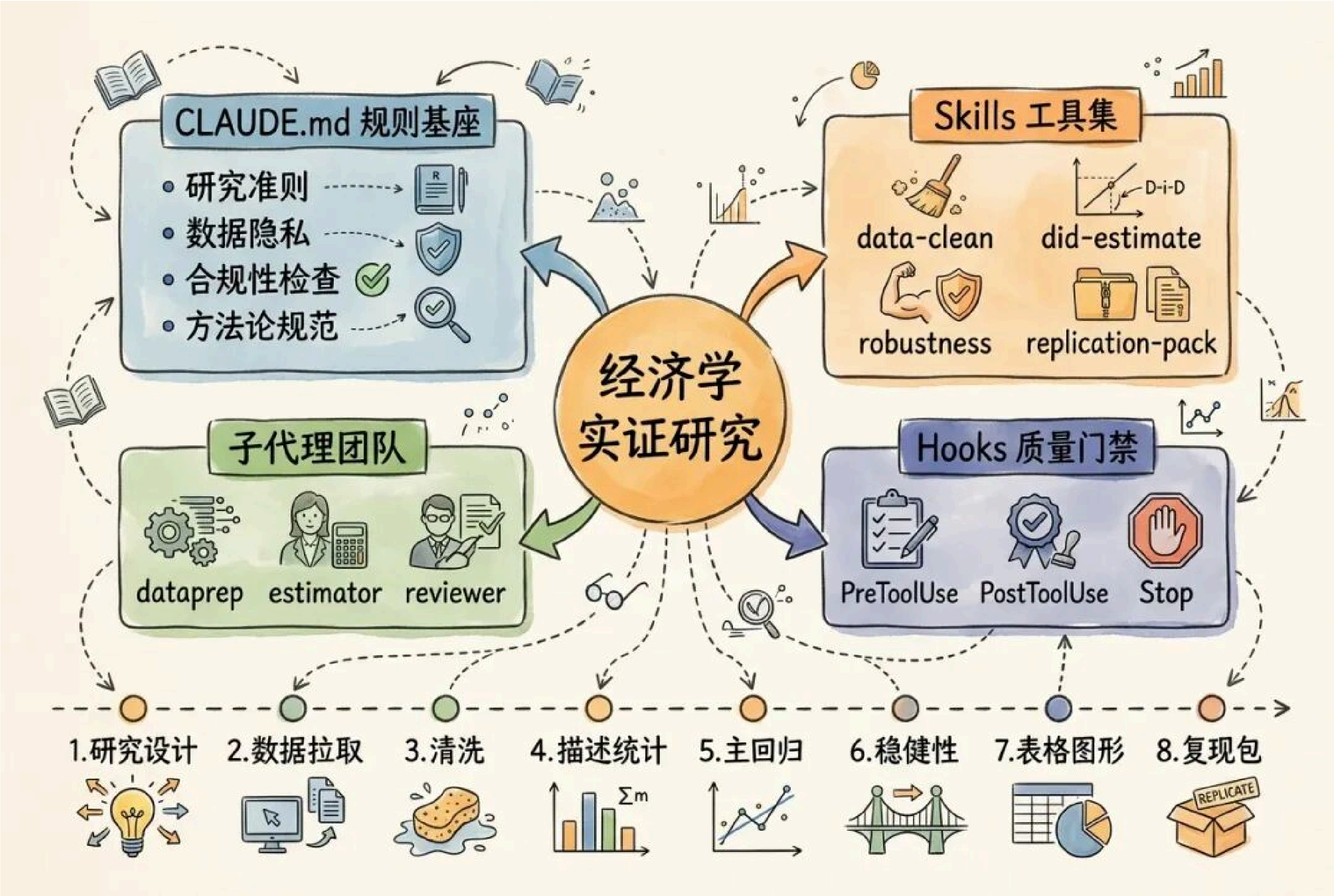
Vibe Coding and the Democratization of Technical Work

Perhaps the most transformative development for researchers is **“vibe coding”**—creating increasingly sophisticated software through natural language descriptions alone. Anthropic’s Claude Code, released in February 2025, exemplifies this breakthrough. OpenAI has since released Codex, which exhibits with similar capabilities.

As is demonstrated in the paper’s econometric tool example, researchers can now build complete analytical tools from simple descriptions. The system handles everything from file uploads to regression analysis to visualization—all generated in minutes without writing a single line of code.

For a discipline where computational methods are increasingly central, but programming skills remain unevenly distributed, this represents a potential leveling of the technical playing field.

The practical implications are clear: economists should engage with these tools now, not merely for productivity gains but to understand their capabilities and limitations. Building your own agents demystifies the technology and reveals both its power and boundaries. It also allows you to benefit more and more as the technology continues to advance.


CLAUDE.md 规则基座
• 研究准则
• 数据隐私
• 合规性检查
• 方法论规范
Skills 工具集
data-clean
did-estimate
D-i-D
robustness
replication-pack
经济学
实证研究
子代理团队
dataprep
estimator
reviewer
Hooks 质量门禁
PreToolUse
PostToolUse
Stop
1.研究设计
2.数据拉取
3.清洗
4.描述统计
5.主回归
6.稳健性
7.表格图形
8.复现包
REPLICATE

Welcome to Claude Code
CLAUDE
CODE
Press Enter to continue

AI Agent配置 — OpenCode

```
Windows PowerShell
版权所有（C） Microsoft Corporation。保留所有权利。

安装最新的 PowerShell，了解新功能和改进！https://aka.ms/PSWindows

PS C:\Users\zzyna> Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
PS C:\Users\zzyna> Get-ExecutionPolicy
RemoteSigned
PS C:\Users\zzyna> git --version
git version 2.54.0.windows.1
PS C:\Users\zzyna> git config --global user.name "xishanyu2"
PS C:\Users\zzyna> git config --global user.email "zzynankai@outlook.com"
PS C:\Users\zzyna> git config --global --list
user.name=xishanyu2
user.email=zzynankai@outlook.com
```

```
PS C:\Users\zzyna> node --version
v24.15.0
PS C:\Users\zzyna> npm install -g opencode-ai

added 3 packages in 24s
PS C:\Users\zzyna> opencode --version
1.15.0
PS C:\Users\zzyna> code --version
1.120.0
0958016b2af9f09bb4257e0df4a95e2f90590f9f
x64
```

```
(base) PS C:\Users\zzyna> pandoc --version
pandoc 3.8
Features: +server +lua
Scripting engine: Lua 5.4
User data directory: C:\Users\zzyna\AppData\Roaming\pandoc
Copyright (C) 2006-2025 John MacFarlane. Web:  https://pandoc.org
This is free software; see the source for copying conditions. There is no
warranty, not even for merchantability or fitness for a particular purpose.
```

这一步是在Anaconda Prompt里

```
(base) C:\Users\zzyna>conda init powershell
no change     D:\anaconda3\Scripts\conda.exe
no change     D:\anaconda3\Scripts\conda-env.exe
no change     D:\anaconda3\Scripts\conda-script.py
no change     D:\anaconda3\Scripts\conda-env-script.py
no change     D:\anaconda3\condabin\conda.bat
no change     D:\anaconda3\Library\bin\conda.bat
no change     D:\anaconda3\condabin\_conda_activate.bat
no change     D:\anaconda3\condabin\rename_tmp.bat
no change     D:\anaconda3\condabin\conda_auto_activate.bat
no change     D:\anaconda3\condabin\conda_hook.bat
no change     D:\anaconda3\Scripts\activate.bat
no change     D:\anaconda3\condabin\activate.bat
no change     D:\anaconda3\condabin\deactivate.bat
modified      D:\anaconda3\Scripts\activate
modified      D:\anaconda3\Scripts\deactivate
modified      D:\anaconda3\etc\profile.d\conda.sh
modified      D:\anaconda3\etc\fish\conf.d\conda.fish
no change     D:\anaconda3\shell\condabin\Conda.psm1
modified      D:\anaconda3\shell\condabin\conda-hook.ps1
no change     D:\anaconda3\Lib\site-packages\xontrib\conda.xsh
modified      D:\anaconda3\etc\profile.d\conda.csh
modified      C:\Users\zzyna\Documents\WindowsPowerShell\profile.ps1

==> For changes to take effect, close and re-open your current shell. <==


(base) C:\Users\zzyna>conda --version
conda 25.11.1

(base) C:\Users\zzyna>python --version
Python 3.13.9
```

```
(base) PS C:\Users\zzyna> git --version
git version 2.54.0.windows.1
(base) PS C:\Users\zzyna> git config --global user.name
xishanyu2
(base) PS C:\Users\zzyna> node --version
v24.15.0
(base) PS C:\Users\zzyna> npm --version
11.12.1
(base) PS C:\Users\zzyna> opencode --version
1.15.0
(base) PS C:\Users\zzyna> code --version
1.120.0
0958016b2af9f09bb4257e0df4a95e2f90590f9f
x64
(base) PS C:\Users\zzyna> conda --version
conda 25.11.1
(base) PS C:\Users\zzyna> pandoc --version
pandoc 3.8
Features: +server +lua
Scripting engine: Lua 5.4
User data directory: C:\Users\zzyna\AppData\Roaming\pandoc
Copyright (C) 2006-2025 John MacFarlane. Web:  https://pandoc.org
This is free software; see the source for copying conditions. There is no
warranty, not even for merchantability or fitness for a particular purpose.
```

```
(base) PS C:\Users\zzyna> mkdir $env:USERPROFILE\opencode-lab


    目录: C:\Users\zzyna


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2026/5/16     21:06                opencode-lab


(base) PS C:\Users\zzyna> cd $env:USERPROFILE\opencode-lab
(base) PS C:\Users\zzyna\opencode-lab> opencode
```

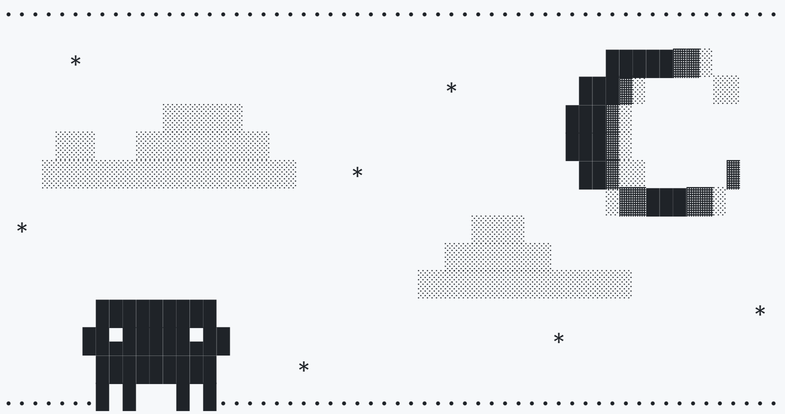
AI Agent配置 — Claude Desktop

- GitHub - farion1231/cc-switch: A cross-platform desktop All-in-One assistant for Claude Code, Codex, OpenCode, OpenClaw, Gemini CLI & Hermes Agent. Only official website: ccswitch.io · GitHub
- 首次调用 API | DeepSeek API Docs
- Xiaomi MiMo API Open Platform
- 智谱AI开放平台、MiniMax...

AI Agent配置 — Claude Code CLI

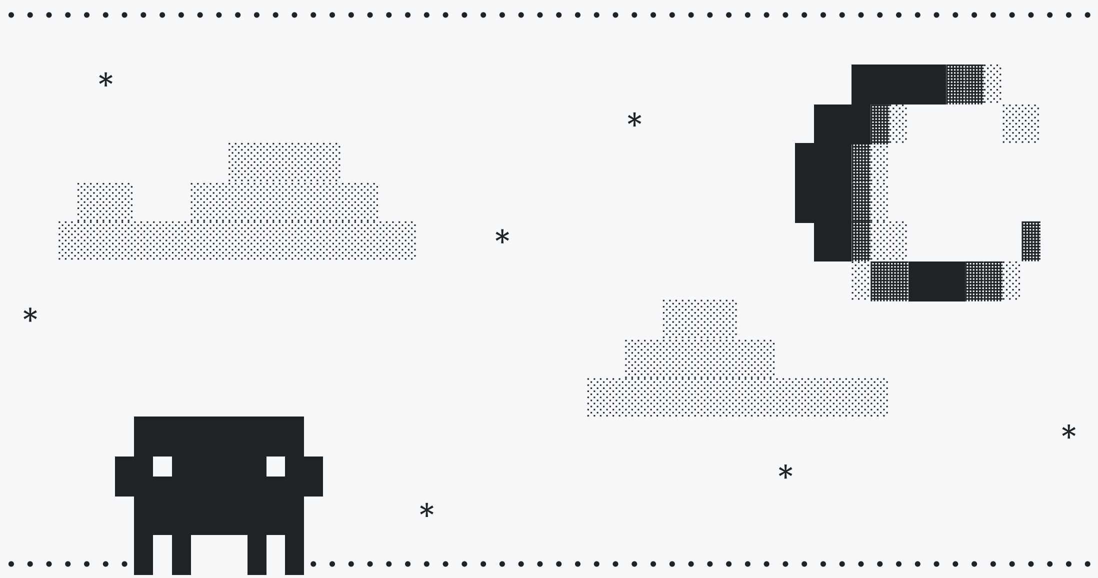
```
(base) PS C:\Users\zzyna> irm https://claude.ai/install.ps1 | iex
```

```
(base) PS C:\Users\zzyna> claude
Welcome to Claude Code v2.1.170
```



```
 Let's get started.

 Choose the text style that looks best with your terminal
 To change this later, run /theme

   1. Auto (match terminal)
 > 2. Dark mode √
   2. Light mode
   3. Dark mode (colorblind-friendly)
   4. Light mode (colorblind-friendly)
   5. Dark mode (ANSI colors only)
   6. Light mode (ANSI colors only)

 ----------------------------------------------------------------------------------------------------------------------
  1  function greet() {
  2 -  console.log("Hello, World!");
  2 +  console.log("Hello, Claude!");
  3  }
 ----------------------------------------------------------------------------------------------------------------------
  Syntax theme: Monokai Extended (ctrl+t to disable)
```

Welcome to Claude Code v2.1.170



Security notes:

1. Claude can make mistakes.
   You're responsible for Claude's actions and should always
   review them, especially when running code.

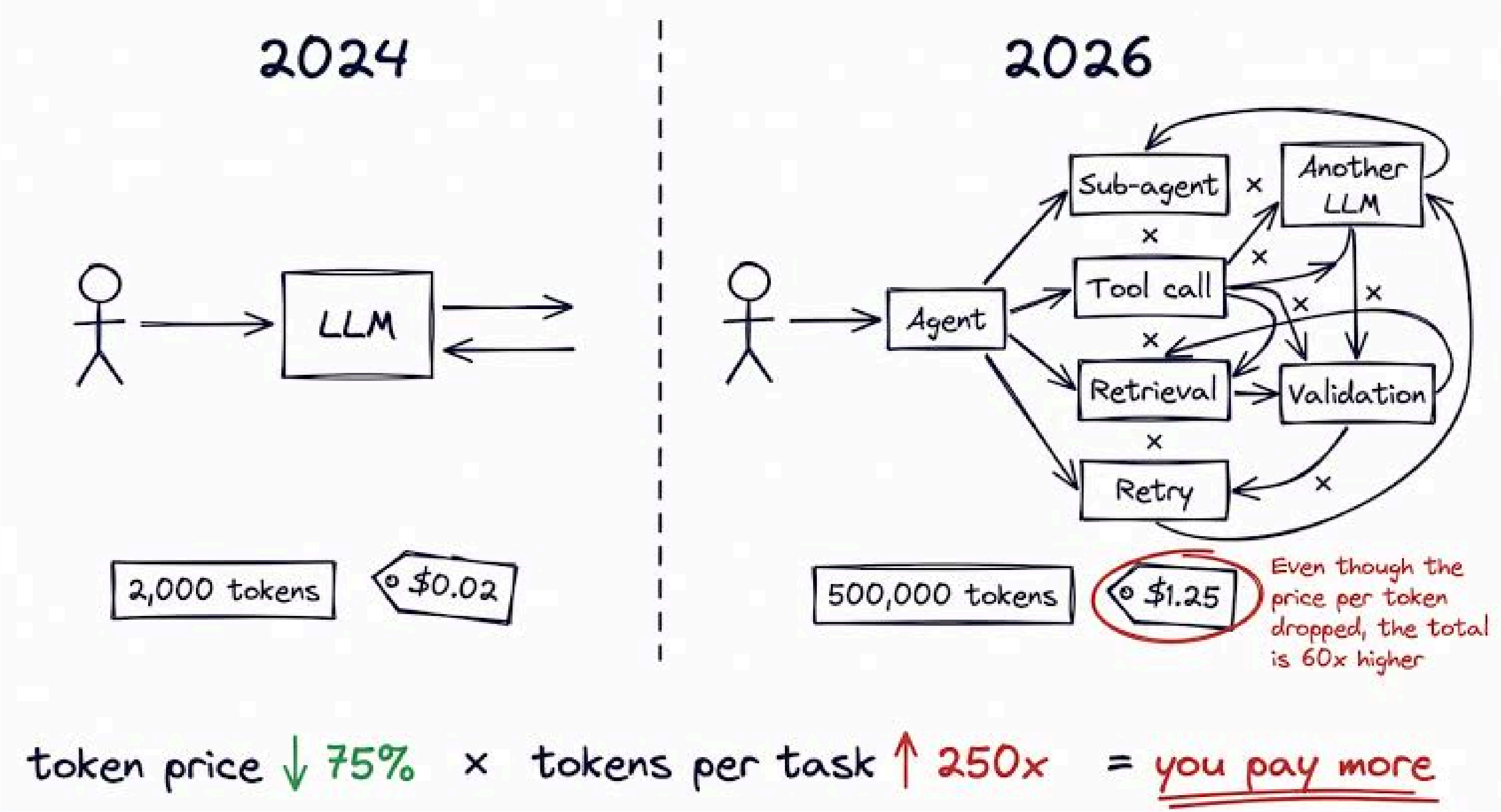
2. Due to prompt injection risks, only use it with code you trust
   Learn more: https://code.claude.com/docs/en/security

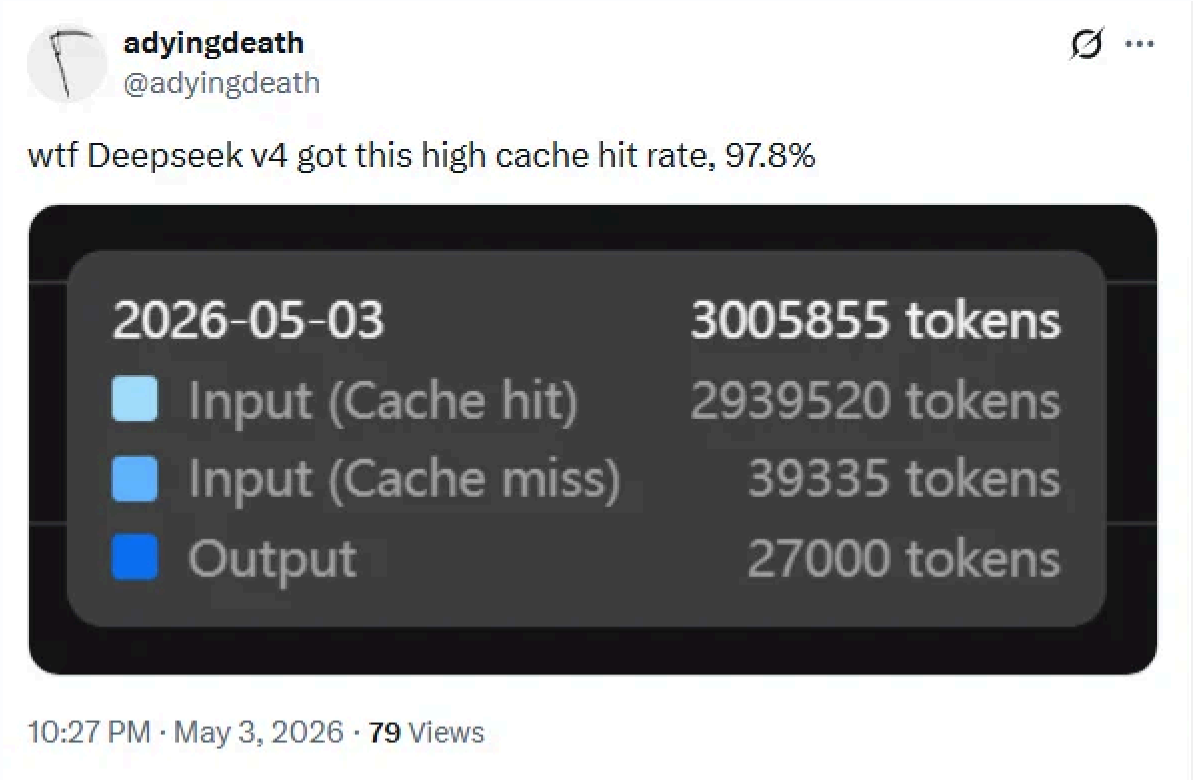
Press Enter to continue…

```
Accessing workspace:

C:\Users\zzyna

Quick safety check: Is this a project you created or one you trust? (Like your own code, a well-known open source
project, or work from your team). If not, take a moment to review what's in this folder first.

Claude Code'll be able to read, edit, and execute files here.

Security guide

> 1. Yes, I trust this folder
  2. No, exit

Enter to confirm · Esc to cancel
```

```
Claude Code v2.1.170

Welcome back!                    | Tips for getting started
                                 | Run /init to create a CLAUDE.md file with instructions for Claude
                                 | Note: You have launched claude in your home directory. For the best experience, l…
                                 | ─────────────────────────────────────────────
                                 | What's new
                                 | Added `--safe-mode` flag (and `CLAUDE_CODE_SAFE_MODE`) to start Claude Code with …
                                 | Added `/cd` command to move a session to a new working directory without breaking…
glm-5.1 · API Usage Billing      | Added a `disableBundledSkills` setting and `CLAUDE_CODE_DISABLE_BUNDLED_SKILLS` e…
C:\Users\zzyna                   | /release-notes for more

>

? for shortcuts · ← for agents                                              ● high · /effort
```

Is AI really getting cheaper?





刚刚，DeepSeek融资差不多落定了，首轮500亿！DeepSeek缓存命中率冲到98%?

03

AI赋能的未来

老子的《道德经》里说，万物负阴而抱阳，冲气以为和。而人类与AI，或许就会在相互的影响中，形成新的和谐体。AI，可能是我们的对手，但它，也能是我们的助手。

人工智能哲学

人工智能焦虑（AI Anxiety）

人工智能迷惘（AI Disarray）

Jensen Huang says it doesn't matter what kids study in the AI era

Parents shouldn't obsess over what their kids study in the era of artificial intelligence, said Nvidia CEO Jensen Huang.

"I think that it won't matter. All the things that used to matter are still things that are going to matter in the future," Huang told Singapore's Channel NewsAsia on Monday.

Instead of chasing AI-proof subjects, **students should focus on using AI to deepen their learning and improve their craft**, Huang said.

The Nvidia chief pointed to journalism, **storytelling, the arts, and design as examples of fields that will remain valuable even as AI becomes more powerful.** He pointed out **the best interviewers are not just well prepared, but able to stay present, listen closely, and respond dynamically in the moment.**

"The ability to tell a story for an audience will remain just as important in the future as it is today," Huang said.

The Nvidia chief also referenced the Japanese concept of "wabi-sabi," or the beauty of imperfection, suggesting that uniquely human qualities could become even more prized in an AI-saturated world.

"Whatever you decide is your passion, the only one thing that you have to do is to make sure that you ask yourself: How can AI help elevate my learning, my craft, my purpose?" he said.

Huang is the latest business leader to weigh in on how AI could reshape education and work.

Earlier this month, futurist and entrepreneur Peter Diamandis told Business Insider that **kids will need qualities such as curiosity, purpose, and adaptability to succeed in the AI era.**

Meanwhile, entrepreneur and entrepreneur-turned-professor Scott Galloway said on "The Diary of a CEO" podcast that parents should focus on helping children **develop durable human skills such as storytelling, communication, and relationship-building.**

Huang echoed those themes, arguing that while AI would automate parts of many jobs, **it would also push people toward higher-level work requiring judgment and creativity.**

"A job is like a basket of tasks," he told CNA. "Many of those tasks will be automated. And my sense is that as a result of automation, **we can focus on the harder parts of our work**."

Huang also pushed back on concerns that widespread AI use could make people less intelligent or lazier thinkers.

Drawing comparisons to the rise of personal computers, the internet, and smartphones, Huang argued that previous waves of technology increased human ambition rather than diminishing it.

"Do we find ourselves busier or less busy? I think the answer is we found ourselves busier," Huang said.

AI的负外部性与监管责任

Dangers of Artificial Intelligence

- Automation-spurred job loss
- Deepfakes and social manipulation
- Privacy violations
- Algorithmic bias caused by bad data
- Socioeconomic inequality
- Weapons and military automatization
- Market volatility
- Increased criminal activity and child safety risks
- Psychological harm and overreliance

课程结语

是结束，也是开始：

知识的边界在哪里？

AI的能力还有多少？

人类的上限在何处？

不要丧失对世界的好奇

不要停止对真理的追求

千里之行，始于足下...